

oscNext L4 — weekly update

Talk script

Recap

Last week I gave a general overview of oscNext, and I also showed a noise BDT. I trained it on pass3 data, using the L4 event selection that I rewrote myself. These are the plots you see here.

Since the selection is my own rewrite, Jan suggested that we run the same code on pass2 and check it against the results that already exist there. So that is what I want to show you today.

Variables

This slide is from last week, and it lists the variables that go into the two BDTs. I do not have the comparison on a slide, so let me just tell you the result.

I rewrote several of these variables myself, because the original L4 script depends on three old projects, and those projects do not exist in our meta-project. So I could not simply run it.

Then I took every variable on this list and compared it with the version already stored in pass2, event by event. The noise inputs were already correct, but the muon inputs were not, so that is where the work went. In the end every variable agrees bitwise, with one exception.

`accumulated_time` agrees in 99.84% of the events, and the largest difference is about one pulse spacing. The original code sorts the pulses with `std::sort`, which is not stable, so pulses that share a time end up in an undefined order. I tried three different tie-break rules and none of them reaches 100%, so I kept the stable sort instead of tuning to it.

Training the noise BDT on pass2

Then I trained the same noise BDT on pass2. Last week I showed the pass3 training, and there were not enough noise events in it, because the pass3 noise simulation is not produced yet. The pass2 noise sample is about sixty times larger.

On the left you see the cut we apply. It rejects 99% of the noise and keeps 96% of the neutrinos. And on the right you see the target from the reference document, which we reach.

So this tells us two things. First, we reproduce the original work. And second, it validates the pass3 results I presented last week.

The score distributions

These plots show the separation, and the noise BDT separates noise from signal cleanly. The left one is linear scale and the right one is log.

The same plot also tells us that the model is not overtrained. The points are the test set, and our criterion is simply that there is no difference between the test set and the training set.

Next

Next comes the muon BDT training. And once everything is done, I move the scripts I wrote into the new repository that Jan set up.

Numbers, if asked

Verification	14 of 15 rows bitwise identical, 56,301 events, 5 samples
Noise events, pass3 to pass2	2,051 to 120,125 (test background 1,016 to 60,504)
Efficiency at 99% rejection	96.9% (pass3: 95.9%, on 10 background events; now 605)
At the production cut of 0.70	about 99.25% rejection, about 96.4% efficiency
Reference target (Table 13)	~96% at 99.2% rejection
accumulated_time residual	99.84% agreement, max difference 286 ns

The overtraining gap for pass2 is read off the --- Overtraining --- block of the training output. On pass3 it was +0.1 points.